Fiscal Year:	FY 2017	**Task Last Updated:**	FY 03/30/2018
PI Name:	Wyatt, Sarah E Ph.D.		
Project Title:	Proteomics Analysis of Arabidopsis Seedlings in Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	45701	**Congressional District:**	15
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	09/01/2013	**End Date:**	08/31/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
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Flight Program:	ISS		
Flight Assignment:	ISS Rapid Turn Around Flight NOTE: End date is now 8/31/2017 per NSSC information and PI (Ed., 10/3/16) NOTE: End date is now 8/31/2016 per NSSC information and PI (Ed., 10/8/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Luesse, Darron Ph.D. (Southern Illinois University, Edwardsville)		
Grant/Contract No.:	NNX13AM48G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The space flight environment provides a unique environment to understand how gravity informs plant growth, development, and physiological processes. The central objective of this study is to determine what proteins are deferentially expressed during space flight in developing Arabidopsis seedlings. A body of research has evaluated the differential gene expression on space flight vs ground controls, but gene expression provides an indirect measure of what proteins are expressed. Genes may be transcribed but the transcripts degraded, or translated proteins may be regulated through post translational modifications. We used BRIC-PDFU (Biological Research In Canisters - Petri Dish Fixation Unit) hardware on a Rapid Turn-Around Space Flight Experiment to gain insights into differences in protein profiles between Arabidopsis seedlings grown during space flight and ground based controls. Arabidopsis seedlings were germinated, grown, and tissues fixed on orbit. Once returned to Earth, total protein was extracted, labeled with iTRAQ reagents, and analyzed for differences in protein profiles, both expression levels and post translational modifications. These data complement the gene expression data currently available from space flight experiments and also provide novel insights into differences in post-transcriptional regulation. They have also provided a foundation for a network analysis to generate additional hypotheses as to the mechanisms involved in gravity perception/response in plants.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research may lead to the discovery of novel components that determine how plants respond to gravity and to the space flight environment. It will also form a foundation for the establishment of hypotheses to support research on plant improvement for space flight and life support in the space flight environment.
Task Progress:	We have now finished all extractions and sequencing and have made substantial progress in data analysis. Just to reiterate, roughly 20,000 Arabidopsis Wild Type (WT) Col-0 seeds were flown aboard the International Space Station (ISS) using PDFU-BRIC hardware. After approximately 3 days, seedlings were fixed using RNAlater® and frozen at -80oC until return to Earth. A duplicate set of WT Col-0 seeds was maintained at Kennedy Space Center, also in PDFU-BRIC hardware, as a ground control. Two additional sets of seedlings, not in PDFU-BRIC hardware but in enclosed containers, were maintained at Ohio University under similar environment conditions as hardware and fixative controls for the space flight samples. Proteins were analyzed using iTRAQ 8-plex tandem mass spectrometry at Danforth Plant Science Center, St. Louis, MO. RNAseq was performed at Washington University, St. Louis, MO. RNA. In space flight and ground controls, 2641 soluble and 2917 membrane proteins were identified using Mascot software (Matrix Biosciences) and the TAIR10 database. The fixative, RNALater, had no detrimental effect on protein quality or identification. Analysis of hardware controls revealed 1651 differentially expressed transcripts, 108 differentially abundant soluble, and 39 membrane proteins. Analysis of samples preserved with and without RNAlater revealed 2750 genes differentially expressed and 84 soluble and 120 membrane proteins. Fold enrichment analysis of fixative controls show evidence of desiccation stress on the transcriptome. Filtering the space flight data for those transcripts and proteins expressed in fixative and hardware controls revealed 571 transcripts, 59 soluble proteins, and 36 membrane proteins deferentially expressed only as a result of space flight. Further analysis will help reveal the processes/mechanisms affected by space flight.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Meyers AD, Kruse CPS, Basu P, Luesse DR, Wyatt SE. "BRIC 20: Integrating omics data of space-flown Arabidopsis." Presented at 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Kruse CPS, Wyatt SE. "At the Intersection: Identification and Characterization of Fundamental Gravity Response Genes." Plant Biology 2017, Honolulu, Hawaii, June 24-28, 2017. Plant Biology 2017, Honolulu, Hawaii, June 24-28, 2017. , Jun-2017
Abstracts for Journals and Proceedings	Meyers AD, Kruse CPS, Basu P, Luesse DR, Wyatt SE. "Arabidopsis Microgravity Omics Data Offer Fresh Perspective on Signaling and Response." Plant Biology 2017, Honolulu, Hawaii, June 24-28, 2017. Plant Biology 2017, Honolulu, Hawaii, June 24-28, 2017. , Jun-2017
Abstracts for Journals and Proceedings	Basu P, Luesse DR, Wyatt SE. "A Transcriptomic and Proteomic Comparison of the Effect of Spaceflight Hardware on Arabidopsis Seedlings." 2017 American Society of Plant Biologists (ASPB) Midwestern Section Annual Meeting, Purdue University, West Lafayette, IN, February 4-5, 2017. 2017 American Society of Plant Biologists (ASPB) Midwestern Section Annual Meeting, Purdue University, West Lafayette, IN, February 4-5, 2017. , Feb-2017
Abstracts for Journals and Proceedings	Wyatt SE. "Arabidopsis Proteomic Results to Microgravity." 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Abstracts for Journals and Proceedings	Basu P, Luesse DR, Wyatt SE. "BRIC 20: Results from proteomic analysis of Arabidopsis seedlings germinated on ISS." 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Articles in Peer-reviewed Journals	Kruse CPS, Meyers AD, Basu P, Hutchinson S, Luesse DR, Wyatt SE. "Spaceflight induces novel regulatory responses in Arabidopsis seedling as revealed by combined proteomic and transcriptomic analyses." BMC Plant Biol. 2020 May 27;20(1):237. https://doi.org/10.1186/s12870-020-02392-6 ; PMID: 32460700; PMCID: PMC7251690 , May-2020

Articles in Peer-reviewed Journals	Basu P, Kruse CPS, Luesse DR, Wyatt SE. "Growth in spaceflight hardware results in alterations to the transcriptome and proteome." Life Sci Space Res (Amst). 2017 Nov;15:88-96. Epub 2017 Sep 21. https://doi.org/10.1016/j.lssr.2017.09.001 ; PubMed PMID: 29198318 , Nov-2017
Articles in Peer-reviewed Journals	Kruse CPS, Basu P, Luesse DR, Wyatt SE. "Transcriptome and proteome responses in RNAlater preserved tissue of Arabidopsis thaliana." PLoS One. 2017 Apr 19;12(4):e0175943. eCollection 2017. https://doi.org/10.1371/journal.pone.0175943 ; PubMed PMID: 28423006; PubMed Central PMCID: PMC5397022 , Apr-2017
Articles in Peer-reviewed Journals	Hutchinson S, Basur P, Wyatt SE, Luesse DR. "Methods for on-orbit germination of Arabidopsis thaliana for proteomic analysis." Gravit Space Res. 2016 Dec;4(2):20-7. http://gravitationalandspacebiology.org/index.php/journal/article/view/743 , Dec-2016
Articles in Peer-reviewed Journals	Kruse CPS, Wyatt SE. "Nitric oxide, gravity response, and a unified schematic of plant signaling." Plant Sci. 2022 Jan:314:111105. https://doi.org/10.1016/j.plantsci.2021.111105 ; PMID: 34895542 , Jan-2022
Books/Book Chapters	Basu P, Kruse CPS, Luesse DR, Wyatt SE. "Plant proteomic data acquisition and data analyses: Lessons from spaceflight." in "Plant Gravitropism. Methods in Molecular Biology" Ed. E.B. Blancaflor. New York, NY: Humana, 2022. vol. 2368: pp. 199-214. https://doi.org/10.1007/978-1-0716-1677-2_13 ; PMID: 34647257, Jan-2022